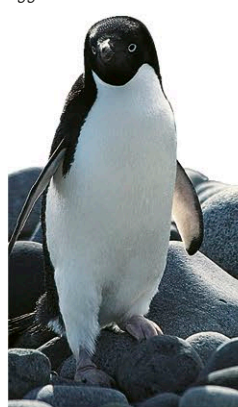
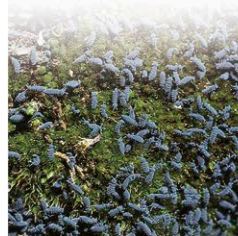




SKUAS, found in both the Arctic and the Antarctic, feed on carrion and on other birds' eggs and chicks.



NESTING COLONIES of some birds, such as Adeline penguins, can contain over a million members.



SPRINGTAILS eat decaying plant remains—unlike most Antarctic animals, which rely on food from the sea.



KRILL harvest plankton, forming the basis of many Antarctic food chains.



LEOPARD SEALS feed on krill, fish, and seabirds. This predatory lifestyle has no direct equivalent in the Arctic.



BALEEN WHALES play a major part in the Southern Ocean's ecology by eating vast amounts of plankton.

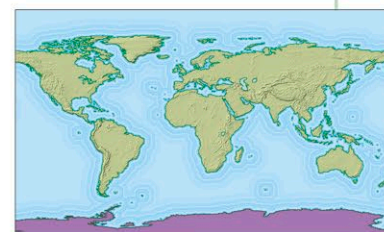
Antarctic

Unlike the Arctic, mainland Antarctica is isolated from the rest of the world. It is covered with ice, up to 13,200 ft (4,000 m) thick, which continues out to sea forming large ice shelves. On the Antarctic Peninsula—a finger of land pointing toward South America—summer temperatures rise to a few degrees above freezing point, but in the rest of the continent, average temperatures are below freezing all year-round.

Algae and lichens grow on bare rocks in many parts of the Antarctic coastline, but the Antarctic Peninsula is the only part of the continent where terrestrial plants can survive. This is also the only place that has a significant range of terrestrial animals, although these are chiefly springtails, mites, and nematode worms—few of which are over 1/8 in (5 mm) long. The rest of

Antarctica's land-based animal life consists of species that feed in the sea and come ashore to breed, such as penguins, or those that scavenge food at these animals' breeding grounds, such as skuas. With the exception of emperor penguins, vertebrates desert the ice at the end of summer to spend the winter at sea.

The Southern Ocean, which surrounds Antarctica, is one of the most biologically productive seas in the world. Although species numbers are relatively low, population sizes are often enormous because the nonstop summer daylight generates a vast food supply. Krill—small crustaceans that form the diet of seals and whales—are especially prolific: some of their swarms are estimated to weigh in excess of 10 million tons and are large enough to be seen by satellites in space. Although the Southern Ocean is always cold, it maintains a minimum temperature of about 28.8°F (-1.8°C); below this, sea water freezes. As a result, the ocean is quite warm compared with Antarctica itself.



MOST OF ANTARCTICA, excluding the relatively mild Antarctic Peninsula, lies south of the Antarctic Circle.

CLIMATE CHANGE

The polar regions are uniquely sensitive to climate change, because global warming—or cooling—has a direct effect on their ice cover, both on land and at sea. Sea ice is rarely more than 16½ ft (5 m) thick, and spreads and retreats with the seasons. It is a vital platform for "ice seals," the approximately 10 species that haul out onto ice to breed,

and it is also used by polar bears and some penguins to get near their food.



GAPS IN THE ICE

The Arctic's summer sea ice is currently shrinking fast, and may disappear altogether by the middle of the 21st century. Paradoxically, Antarctica's sea ice is slowly increasing—a trend that may be linked to changes in wind patterns.